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A “Smith’s Prize” Paper; Solutions (continued).

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whence, elements of area are

$$dc \sin c \, dl, \quad dc' c' \, dl',$$

tangents of azimuth are

$$\frac{dc}{\sin c \, dl}, \quad \frac{dc'}{c' \, dl'}.$$

and substituting the values $l' = l$, $c' = f(c)$, we have $dl' = dl$, $dc' = f'(c)dc$; and thence elements of area are as $\sin c : f(c)f'(c)$,

$$\text{tangents of azimuth as } \frac{dc}{\sin c \, dl} : \frac{f'(c)dc}{f(c) \, dl}.$$

By proper determinations of the function $f(c)$, we can make

- (1) The ratio of the elements of area to be constant; this will be the case if

$$f(c)f'(c) = k \sin c, \quad \text{that is, } f^2(c) = \text{const.} - 2k \cos c,$$

since $f(c)$, $= c$, must vanish for $c = 0$; that is,

$$f^2(c) = 2k(1 - \cos c);$$

- (2) corresponding azimuths to be equal; this will be the case if

$$\frac{f'(c)}{f(c)} = \frac{1}{\sin c},$$

or say

$$f(c) = k \tan \frac{1}{2}c.$$

This is in fact the stereographic projection, in which (as is known) any indefinitely small figure is in the map represented without distortion.

4. *Explain the general configuration of the contour and slope lines in a tract of Lake and Mountain country.*

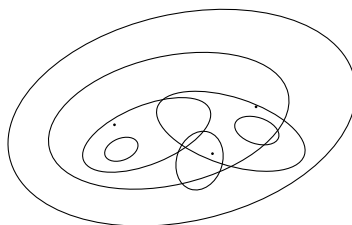
A contour line is the locus of points having a given altitude.

To fix the ideas, consider an island forming a two-headed mountain. The contour line at the sea level is a closed curve; at a sufficiently great altitude the contour line consists of two closed curves surrounding the two summits respectively; the transition from one form to the other takes place at the altitude of the pass between the two summits, the contour line then having a node at the top of the pass, and being in form a figure of eight. At the altitude of the lower summit one of the closed curves is reduced to a point, and for greater altitudes it disappears, the contour line being then a single closed curve surrounding the higher summit; and at the altitude of the higher summit this reduces itself to a point. (See fig. 1.) If there is on the breast of the mountain a lake, the contour line at the lake level is (as in

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the case of the pass) a curve with a node, having however here a closed curve with an interior loop as shown in the figure. The contour line for an altitude below the lake level includes as part of itself a closed curve lying within the contour of the lake,

FIG. 1.



and which for the altitude of the lowest point (or “imit”) of the lake reduces itself to a point, and for smaller altitudes disappears.

The contour lines in the immediate neighbourhood of a summit are in general ellipses (geometrically, the indicatrix is an ellipse); they may however be circles (viz. if the summit be an umbilicus).

A slope line is a line of greatest inclination, and it is consequently an orthogonal trajectory to the series of contour lines. A slope line may be considered as always terminated in a summit or an imit; there is through each summit (or imit) an infinity of slope lines, and in general these all touch there; if, however, the contour lines in the immediate neighbourhood are circles, then the slope lines, instead of touching, pass from the point in all directions. Through the node at the top of a pass, or outlet of a lake, there are two intersecting slope lines, one ascending each way from the node, and being in general a “ridge-line”; the other descending each way, and being in general a “course-line,” viz. in the case of a pass, it is in each direction the course of the principal stream of the valley; and in the case of a lake-outlet, it is in the direction away from the lake, the course of the out-flowing stream. The slope lines which thus pass through the several nodes mark out distinct regions, and so facilitate the tracing of the intermediate slope lines.

5. Find the differential equations corresponding to the three integral equations respectively (i) $(y + c)^2 = x(x - 1)(x - 2)$; (ii) $(y + c)^2 = x^2(x - 1)$; and (iii) $(y + c)^2 = x^3$; and discuss geometrically the singular solutions.

Generally, for the equation $(y + c)^2 - X = 0$, the derived equation is

$$X'^2 \left(\frac{dx}{dy} \right)^2 - 4X = 0.$$

If from the integral equation, differentiating in regard to c , we attempt to find the singular solution, we obtain $(y + c) = 0$; and thence $X = 0$ for the singular solution. It is however to be observed that, if X contain single and multiple factors,

$$X = (x + \alpha)(x + \beta) \dots (x + \gamma)^m \dots,$$

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then it is only the single factors $x + \alpha = 0$, $x + \beta = 0$, $\dots$, which are solutions of the differential equation when expressed in its proper form free from extraneous factors.

To explain how this is, write the differential equation in the form

$$X'^2 \left(\frac{dx}{dy} \right)^2 - 4X = 0;$$

for a single factor $x + \alpha$, X' does not contain the factor $x + \alpha$, and there is no division by $x + \alpha$. The equation $x + \alpha = 0$ gives $\frac{dx}{dy} = 0$, $X = 0$, and the equation is thus satisfied; and by what precedes $x + \alpha = 0$ is a singular solution. Contrariwise, for the multiple factor $(x + \gamma)^m$, X' will contain $(x + \gamma)^{m-1}$, and the equation $X'^2 \left(\frac{dx}{dy} \right)^2 - 4X = 0$ will divide by $(x + \gamma)^m$, and divested of this factor it will be of the form

$$\frac{X'^2}{(x + \gamma)^m} \left(\frac{dx}{dy} \right)^2 - \frac{4X}{(x + \gamma)^m} = 0,$$

where

$$\frac{X'^2}{(x + \gamma)^m}$$

contains the factor $(x + \gamma)^{m-2}$ (index is 0 or positive) but

$$\frac{X}{(x + \gamma)^m}$$

does not contain $x + \gamma$; hence the equation $x + \gamma = 0$ gives $\frac{dx}{dy} = 0$, and therefore

$$\frac{X'^2}{(x + \gamma)^m} \left(\frac{dx}{dy} \right)^2 = 0,$$

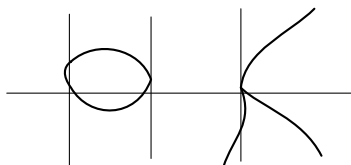
but it does not give $\frac{X}{(x + \gamma)^m} = 0$, and consequently fails to satisfy the differential equation.

Reverting to the integral equation $(y + c)^2 = (x + \alpha)(x + \beta) \dots (x + \gamma)^m \dots$, we see that $x + \alpha = 0$ touches each of the series of curves, and is thus an envelope thereof; that $x + \gamma = 0$ is not an envelope, but is the locus of a singular point on the series of curves.

Applying the foregoing considerations to the proposed equation,

(i) The differential equation is (fig. 2),

FIG. 2.



$$(3x^2 - 6x + 2)^2 \left(\frac{dx}{dy} \right)^2 - 4x(x - 1)(x - 2) = 0,$$

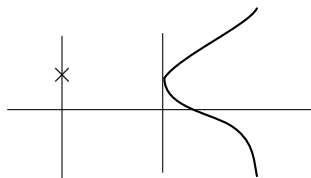
having the singular solutions $x = 0$, $x - 1 = 0$, $x - 2 = 0$.

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(ii) The differential equation is (fig. 3),

$$(3x - 2)^2 \left(\frac{dx}{dy} \right)^2 - 4(x - 1) = 0,$$

FIG. 3.



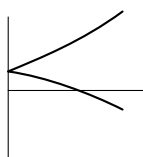
having the singular solution $x - 1 = 0$. But $x = 0$ is not a solution; it is the locus of the series of conjugate points of the curves $(y + c)^2 = x^2(x - 1)$.

(iii) The differential equation is (fig. 4),

$$9x \left(\frac{dx}{dy} \right)^2 - 4 = 0,$$

which has no singular solution. But $x = 0$ is the locus of the cusp of the curves $(y + c)^2 = x^3$.

FIG. 4.



6. Show that the curve parallel to the parabola is of the order 6 and class 4; explain the reduction of class; and trace the system of parallel curves.

The curve parallel to the parabola is the envelope of a circle of constant radius, having its centre on a parabola; taking the equation of the parabola to be $y^2 = 4x$, the coordinates of any point thereon may be taken to be $a^2, 2a$, and the equation of the variable circle is

$$(x - a^2)^2 + (y - 2a)^2 = r^2,$$

that is,

$$a^4 + a^2(-2x + 4) + a(-4y) + x^2 + y^2 - r^2 = 0,$$

or multiplying by 6, this is

$$(a, 0, c, d, e)(a, 1)^4 = 0,$$

where

$$a = 6,$$

$$c = -2x + 4,$$

$$d = -6y,$$

$$e = 6(x^2 + y^2 - r^2).$$

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The equation of the envelope is obtained by eliminating a between the equation in a and the derived equation; or, what is the same thing, by equating to zero the discriminant of the equation in a ; we thus obtain

$$(ae + 3c^2)^3 - 27(ace - ad^2 - c^3)^2 = 0,$$

which, substituting for a, c, d, e their values, is an equation in (x, y) of the order 6; the order of the curve is thus = 6.

The class is most easily obtained by geometrical considerations. Seeking the tangents which can be drawn from a given point, it at once appears that, describing about this point a circle radius r , then to each tangent through the point to the parallel curve, there corresponds a common tangent of the circle and the parabola, and reciprocally; whence the class of the curve is equal to the number of common tangents of the circle and the parabola, that is, it is = 4.

To the order 6 corresponds in general a class = 30, and the reduction from 30 to 4, = 26, is caused by the cusps and nodes of the curve.

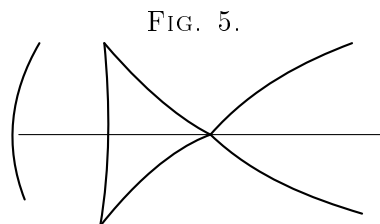
The cusps are given as the points of intersection of the curves $ae + 3c^2 = 0$, $ae - ad^2 - c^3 = 0$, which being respectively of the orders 2 and 3, give 6 cusps; it may be added that these cusps (2 real and 4 imaginary, or else all 6 imaginary) are, in regard to the parabola, the centres of curvature for those points at which the radius of curvature is = r .

There is on the axis a single point (always real) whose normal distance from the parabola is = r . Such point is a node of the parallel curve, viz. according to the value of r , either an acnode (conjugate point) or a crunode: we have thus 1 node. The parallel curve at infinity coincides with the two parabolas obtained by the displacement of the given parabola parallel to itself through the distances $+r, -r$ along the axis of y . Two such parabolas have at infinity on the axis of x a contact of the second order, equivalent to three coincident intersections; and the parallel curve has thus at infinity on the axis of x , a singular point equivalent to 3 nodes; the number of nodes is thus = 4; and the 4 nodes and 6 cusps give the required reduction $2 \cdot 4 + 3 \cdot 6 = 26$.

The parallel curve is most easily traced geometrically; there are two branches, one outside, the other inside the parabola, equidistant from it. The outside branch is a curve of continuous curvature, the form of which requires no explanation. As regards the inside branch, when r is small, this is also of continuous curvature, but there is on the axis of x inside the branch a real acnodal point (the node above referred to): when r becomes = radius of curvature at vertex (or twice the focal distance), the acnode coincides with the branch; and the point on the axis, although presenting no visible singularity, is really in the nature of a triple point composed of two cusps and a node; when r is greater than this limiting value, instead of the acnode we have a crunode;

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and two real cusps present themselves; viz. the form of the inside branch is as shown in fig. 5.



7. *Show that a curve of the order n has at most $\frac{1}{2}(n-1)(n-2)$ double points; and that in any curve having this maximum number of double points, the coordinates (x, y, z) may be taken to be proportional to rational and integral functions of a variable parameter.*

A curve of the order n cannot have more than $\frac{1}{2}(n-1)(n-2)$ double points; for suppose it had one more, say

$$\frac{1}{2}(n-1)(n-2) + 1 = \frac{1}{2}(n^2 - 3n) + 2 \text{ double points;}$$

then since a curve of the order $n-2$ can be drawn through

$$\frac{1}{2}(n-2)(n+1) = \frac{1}{2}(n^2 - n) - 1 \text{ points,}$$

suppose it drawn through the

$$\frac{1}{2}(n^2 - 3n + 2) \text{ double points;}$$

and besides through

$$n-3 \text{ points on the curve,}$$

together

$$\frac{1}{2}(n^2 - n) - 1 \text{ points,}$$

it will cut the curve of the order n in the double points considered as

$$n^2 - 3n + 4 \text{ points,}$$

and besides in

$$n-3 \text{ points,}$$

together

$$n^2 - 2n + 1 \text{ points,}$$

that is, we should have a curve of order $n-2$ meeting the curve of the order n in $n(n-2) + 1$ points, which is of course impossible.

Consider now a curve of the order n with $\frac{1}{2}(n-1)(n-2)$ double points; draw a curve of the order $(n-2)$ through the double points; and besides through

$$\frac{1}{2}(n^2 - 3n + 2) \text{ points,}$$

together

$$n-3 \text{ points on the curve,}$$

together

$$\frac{1}{2}(n^2 - n) - 2 \text{ points,}$$

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this imposes on the curve $\frac{1}{2}(n^2 - n) - 2$ conditions; but as the curve might have been determined to satisfy $\frac{1}{2}(n + 1)(n - 2)$ conditions, this will leave still the curve indeterminate parameter λ . It meets the curve n in the double points considered as

$$n^2 - 3n + 2 \text{ points,}$$

and besides in

$$n - 3 \text{ points,}$$

altogether

$$n^2 - 2n \text{ points;}$$

that is, in

$$2 \text{ points;}$$

the coordinates of which are expressible rationally in terms of λ .

Hence since for any value whatever of λ , there is only one remaining intersection, the coordinates of this point must be rationally determinable in terms of the parameter λ ; that is, the coordinates of any point on the given curve will be rational functions of λ ; or using trilinear coordinates, the coordinates x, y, z of any point on the given curve will be proportional to rational and integral functions of λ .

8. *Show that three bodies attracting each other according to the law of gravitation may move in a line in such wise that the mutual distances are in a constant ratio the value of which depends on the masses.*

Taking the masses m_1, m_2, m_3 , to be arranged in this order at distances z_1, z_2, z_3 , from a fixed origin, the equations of motion are

$$\frac{d^2 z_1}{dt^2} = -\frac{m_2}{(z_1 - z_2)^2} - \frac{m_3}{(z_1 - z_3)^2},$$

$$\frac{d^2 z_2}{dt^2} = \frac{m_1}{(z_1 - z_2)^2} - \frac{m_3}{(z_2 - z_3)^2},$$

$$\frac{d^2 z_3}{dt^2} = \frac{m_1}{(z_1 - z_3)^2} + \frac{m_2}{(z_2 - z_3)^2}.$$

Assume

$$\frac{d^2(z_1 - z_2)}{dt^2} = -\frac{m_1 + m_2}{(z_1 - z_2)^2} - \frac{m_3}{(z_1 - z_3)^2} + \frac{m_3}{(z_2 - z_3)^2},$$

and therefore

$$z_1 - z_2 = z_2 - z_3 = z_1 - z_3,$$

then the equations become

$$\frac{d^2(z_1 - z_2)}{dt^2} = \left\{ -m_1 - m_2 + \frac{m_3}{(\lambda + 1)^2} - \frac{m_3}{\lambda^2} \right\} \frac{1}{(z_1 - z_2)^2},$$

and

$$\frac{d^2(z_1 - z_3)}{dt^2} = \left\{ \frac{m_1}{\lambda^2} - \frac{m_3}{(\lambda + 1)^2} + \frac{m_2}{(\lambda + 1)^2} - m_3 \right\} \frac{1}{(z_1 - z_3)^2},$$

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which equations will be one and the same equation if only

$$\frac{1}{\lambda+1} \left\{ -m_1 - m_2 + \frac{m_3}{(\lambda+1)^2} - \frac{m_3}{\lambda^2} \right\} = \frac{m_1}{\lambda^2} - \frac{m_3}{(\lambda+1)^2} + \frac{m_2}{(\lambda+1)^2} - m_3,$$

an equation which (multiplying out) is of the fifth order, and gives at least one real value of λ ; and λ being thus determined, we may from either equation obtain an equation of the form $\frac{d^2(z_1 - z_2)}{dt^2} = -\frac{M}{(z_1 - z_2)^2}$, which determines $z_1 - z_2$ in terms of t , and then $z_1 - z_2$, $-z_1(z_1 - z_2)$, is also known; that is, the relative motions of the three bodies are determined.

9. *Write down the integral equations for the elliptic motion of a planet; and if r , a , b , denote the radius vector, longest of the latitude, and central longitude respectively, show that*

$$\frac{\sqrt{(1+e)}}{\sqrt{r}} = \frac{1}{a(1-e^2)} \sqrt{(1+e)} + e \cos(s - \omega),$$

explaining the significations of the constants a , e , ω .

Write

a , the mean distance,

n , the mean motion $= \frac{\sqrt{\mu}}{a^{3/2}}$,

e , the eccentricity,

θ , longitude of node,

ω , longitude of perihelion in orbit,

ε , the longitude in orbit.

The position in the orbit is determined in terms of the time by means of an auxiliary quantity u , viz. writing

$$nt + \varepsilon - \omega = u - e \sin u.$$

We then have the radius vector r , and the true anomaly f , given as functions of the time by the equations

$$\cos f = \frac{\cos u - e}{1 - e \cos u},$$

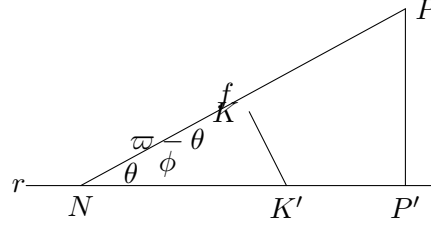
$$\sin f = \frac{\sqrt{(1-e^2)} \sin u}{1 - e \cos u},$$

$$r = \frac{a(1-e^2)}{1+e \cos f} = a(1-e \cos u).$$

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Take s , θ , the hypotenuse, base, and perpendicular of the right-angled triangle SPP' , base angle $\omega - \theta$ (see fig. 6), in which SP is the orbit, S the node, K the perihelion, P the planet; then

FIG. 6.



$\varepsilon = \omega - \theta + f$ gives s ; and so gives in terms of r, θ , by the formulæ relating to the spherical triangle, so that we have

$$\begin{aligned} \text{longitude in orbit} &= s + \theta (= \omega + f), \\ \text{reduced longitude} &= b, \\ \text{latitude} &= p. \end{aligned}$$

The expression for r , writing for f its value, $= \varepsilon - \omega$, gives

$$\frac{1}{r} = \frac{1}{a(1-e^2)} \{1 + e \cos(\varepsilon - \omega)\},$$

and we have thence

$$\frac{\sqrt{(1+e)}}{\sqrt{r}} = \frac{1}{a(1-e^2)} \sqrt{(1+e)\{1 + e \cos(\varepsilon - \omega)\}}.$$

But we have

$$1 - e = 1 - e - e + e \cos(\varepsilon - \omega).$$

And thence

$$\cos(\varepsilon - \omega) = \cos(\varepsilon - \theta) + \sin(\varepsilon - \omega) \sin(\omega - \theta).$$

Consequently

$$\begin{aligned} \sqrt{(1+e) \cos(\varepsilon - \omega)} &= \sqrt{(1+e) \cos(\varepsilon - \theta) \cos(\omega - \theta)} + \sin(\varepsilon - \theta) \sin(\omega - \theta); \\ &= \sqrt{(1+e) \cos(\omega - \theta) \cos(\varepsilon - \theta)} + \sin(\omega - \theta) \sin(\varepsilon - \theta); \end{aligned}$$

but by the right-angled triangle, we have

$$\sin(\varepsilon - \theta) \sin \phi = \sin \omega, \quad = \frac{\sin \omega}{\sqrt{(1+e)}},$$

and thence

$$\begin{aligned} \sin(\omega - \theta) &= \sin \phi \sin \gamma = \tan \phi \cos \theta \tan \theta, \\ \sqrt{(1+e) \cos(\omega - \theta)} &= \frac{1}{\cos \phi} + \cos \phi = \frac{\sin(\varepsilon - \theta)}{\sqrt{(1+e)}}, \\ \cos(\varepsilon - \theta) \cos(\omega - \theta) &= \cos \omega + \frac{\cos \theta \sin(\omega - \theta)}{\sin(1+e)}, \end{aligned}$$

that is,

$$\sqrt{(1+e) \cos(\omega - \theta)} = 1 + e \cos(\omega - \theta).$$

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The formula thus becomes

$$\begin{aligned}\frac{\sqrt{(1+e)}}{\sqrt{r}} &= \frac{1}{a(1-e^2)} \sqrt{(1+e)\{\cos(\varepsilon-\theta)\cos(\omega-\theta) + \sin(\varepsilon-\theta)\sin(\omega-\theta)\}} \\ &= \frac{1}{a(1-e^2)} \sqrt{(1+e)\{1+e\cos(\omega-\theta)\}\cos(\varepsilon-\theta) + e\sin(\omega-\theta)\sin(\varepsilon-\theta)}, \\ &= \frac{1}{a(1-e^2)} \sqrt{(1+e) + e_1\cos(\varepsilon-\theta_1)},\end{aligned}$$

where

$$\frac{1}{\cos\phi} \tan(s-\theta) = \tan(\omega-\theta) \text{ gives } s_1,$$

$$e_1 = \frac{\cos(s-\theta)}{\cos(\omega-\theta)} \text{ gives } e_1,$$

$$e_1(1-e_1^2) = 1+e_1 \text{ gives } a_1.$$

The first equation shows that, drawing the arc KK' at right angles to SP , then

$$\varepsilon_1 = \theta + KK';$$

and therefore

$$m_1 = \theta + \sin\omega, e_1 = e\cos\phi, e_1 = TK'.$$

The other two equations then give e_1 and a_1 ; it may be added that, from the right-angled triangle NKK' we have $\cos(m_1-\theta) = \cos\phi\cos(\omega-\theta)$; and consequently that $a, (1-e) = a, (1-K')$.

10. *Find the differential equations for the motion of a material line acted upon by any forces and moving in a given outer surface.*

If a, b are the coordinates of the point of intersection of any line of the ruled surface with the plane of xy ; α, β, γ the cosines of the inclinations of the line to the three axes respectively; then writing

$$\frac{\beta}{a} = \alpha, \quad b - a\frac{a}{\beta} = b, \quad \frac{\gamma}{\beta} = \gamma,$$

and considering a, b, β, γ as given functions of a variable parameter θ , when a, β, γ satisfy the relation $a^2 + \beta^2 + \gamma^2 = 1$, the equations in question, coordinates of the equation ($= \varepsilon$) determine the position of the line on the surface, and taking account of the equation ($= \varepsilon$), they determine its form of the parameters θ, θ , the coordinates of a particular point in this line.

The motion of the material line is such that it must successively to coincide with the several lines on the ruled surface, consider on the material line a fixed point; say the centre of gravity G , and imagine that in the course of the motion the material line comes to coincide with the line determined as above by the parameter θ , and that the point G with the point determined as above by the parameters τ, θ , consider as

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material line any other point P whose distance from G is $= s$; the parameters of P will be $r + s, \theta$; and the coordinates will consequently be given in terms of the variable parameters r, θ , by the equations

$$\frac{x - a}{\alpha} = \frac{y - b}{\beta} = \frac{z}{\gamma} = r + s,$$

that is, we have

$$x = a + (r + s)\alpha, \quad y = b + (r + s)\beta, \quad z = (r + s)\gamma,$$

where $a, b, \alpha, \beta, \gamma$ are given functions of θ ; r, θ are parameters varying with the time, and s is a constant in regard to the time, but varies with the point under consideration. Hence, writing as usual T to denote the *Vis Viva* function

$$T = \frac{1}{2} \int (x'^2 + y'^2 + z'^2) dm,$$

where dm is the element of the material line, and x', y', z' are the velocities of the element, it only remains to express T as a function of r, θ ; and the equations of motion will be

$$\frac{d}{dt} \frac{\partial T}{\partial r'} - \frac{\partial T}{\partial r} = R, \quad \frac{d}{dt} \frac{\partial T}{\partial \theta'} - \frac{\partial T}{\partial \theta} = \Theta.$$

Using $a', b', \alpha', \beta', \gamma'$ to denote the differential coefficients in regard to θ , but, as above, x', y', z', r', θ' to denote differential coefficients in regard to t , we have

$$\begin{aligned} x' &= \{a' + (r + s)\alpha'\}\theta' + \alpha r', \\ y' &= \{b' + (r + s)\beta'\}\theta' + \beta r', \\ z' &= (r + s)\gamma'\theta' + \gamma r', \end{aligned}$$

and thence

$$x'^2 + y'^2 + z'^2 = A\theta'^2 + 2Br'\theta' + r'^2,$$

if for shortness

$$\begin{aligned} A &= a'^2 + b'^2 + 2(r + s)(\alpha a' + \beta b') + (r + s)^2(\alpha'^2 + \beta'^2 + \gamma'^2), \\ B &= a'\alpha + b'\beta + (r + s)(\alpha\alpha' + \beta\beta' + \gamma\gamma') = a'\alpha + b'\beta, \end{aligned}$$

since

$$\alpha\alpha' + \beta\beta' + \gamma\gamma' = 0.$$

Hence we have

$$T = \frac{1}{2}\theta'^2 \int A dm + r'\theta' \int B dm + \frac{1}{2}r'^2 \int dm.$$

Observing that in the sum $\int A dm$, the terms involving $\int s dm$ vanish in consequence of G being the centre of gravity, we have

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Or writing $\int dm = M$, $\int s^2 dm = Mk^2$, (M being therefore the mass of the line, and Mk^2 its moment about the centre of gravity), we have

$$\int A dm = M\{a'^2 + b'^2 + 2r(\alpha a' + \beta b') + (r^2 + k^2)(\alpha'^2 + \beta'^2 + \gamma'^2)\}.$$

Moreover

$$\int B dm = M(a'\alpha + b'\beta),$$

and hence we have

$$T = \frac{1}{2}M\{\theta'^2[a'^2 + b'^2 + 2r(\alpha a' + \beta b') + (r^2 + k^2)(\alpha'^2 + \beta'^2 + \gamma'^2)] + 2\theta'r'(a'\alpha + b'\beta) + r'^2\},$$

a given function of r, θ, r', θ' ; and the differential equations are therefore given as above.

11. *Explain the mutual connexion of the three theorems in conics: (1) the theorem at quartet linear; (2) Pascal's theorem; (3) the theorem of the anharmonic relations of four points.*

The theorem of quartet linear is that the locus of a point, such that the product of its distances from two given lines is to always in a given ratio to the product of its distances from two other given lines, is a conic.

Consider the lines as forming a quadrilateral $ABCD$. Then A, B, C, D are points on the conic, and writing PAB to denote the perpendicular distance of a point P from the line AB , and so in other cases, the theorem is that the expression

$$\frac{AC \cdot BD \sin PAB \cdot \sin PBD}{AB \cdot DC \sin PAD \cdot \sin PBC}$$

has a constant value for any point P whatever on the conic. Now the perpendicular distance $PAB = 2AB \cdot AB = AB$, or, what is the same thing, it is $= PA \cdot PB \sin PAB \div AB$; transforming the other perpendicular distances in the same manner, the distances PA, PB, PC, PD divide out, and the foregoing expression becomes

$$\frac{AC \cdot BD \sin PAB \cdot \sin PBD}{AB \cdot CD \sin PAD \cdot \sin PBC};$$

viz. omitting the constant factor, it appears that the expression

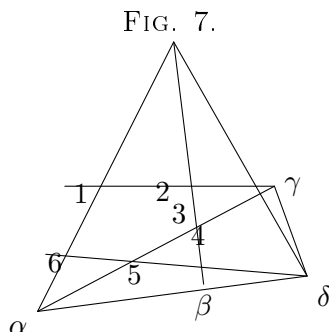
$$\frac{\sin PAB \cdot \sin PCD}{\sin PAD \cdot \sin PBC}$$

constant for all points P on the conic, or, what is the same thing, that the anharmonic ratio of the pencil $P(A, B, C, D)$ is constant for all points P on the conic. This is the anharmonic property of the points of a conic.

Pascal's theorem is that for any six points 1, 2, 3, 4, 5, 6 (fig. 7) on a conic, the intersections of the lines 12 and 45, the lines 23 and 56, the lines 34 and 61 lie in a line. Marking the points $\alpha, \beta, \gamma, \delta$ as shown in the figure, it appears by the theorem that we have the two lines (65 $\beta\delta$), (α 547) meeting in 5, and such that the lines through the corresponding points 6 and α, β and 4, δ and γ meet in a point. Hence the ranges (65 $\beta\delta$) and (α 547) have the same anharmonic ratio; or, what is the same thing, the pencils 3(65 $\beta\delta$) and 1(α 547) have the same anharmonic ratio; that is, the pencils 3(6542) and 1(6542) have the same anharmonic ratio; or, considering 1 as a variable

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point, but 2, 3, 4, 5, 6 are fixed points on the conic, the pencil 1(6542) of lines from the point 1 to the four points 6, 5, 4, 2 has the same anharmonic ratio for all positions whatever of the variable point 1; which is the above-mentioned anharmonic property of the points of a conic.



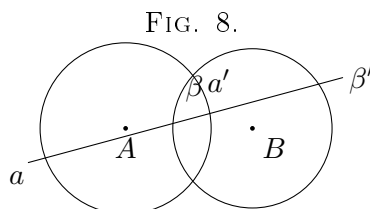
12. Show that any line through the centre of either of two orthotomic circles cuts the two circles harmonically; and connect this result with the theorem that the Jacobian of three circles is made up of the line infinity and the orthotomic circle.

Drawing through the centre of the circle A , (fig. 8) the line $a\beta a'\beta'$, it appears by the figure that we have

$$A\beta \cdot A\beta' = \text{square of tangential distance of } A \text{ from the circle } B,$$

$$= Aa'^2 \quad \text{since the circles cut at right angles,}$$

$$= Aa \cdot Aa';$$



that is, the points β, β' are inverse points in regard to the circle on the diameter aa' , and the points a, a', β, β' are thus harmonically related to each other.

Hence the polar of a , in regard to the circle B , passes through the point a' , which is the opposite of a in regard to the circle A ; and it is consequently a fixed point for all the circles B orthotomic to the circle A . That is, considering any three circles orthotomic to the circle A , this circle A is a locus of points a such that the polars of a in regard to the three circles meet in a point.

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Moreover, all circles meet the line at infinity in two fixed points I, J , the circular points at infinity: hence for any point a on the line infinity, the polar of a in regard to any circle whatever meets the line infinity in a fixed point, the harmonic of a in regard to the two points I, J ; whence the line infinity is also a locus of points a such that the polars of a in regard to the three given circles meet in a point.

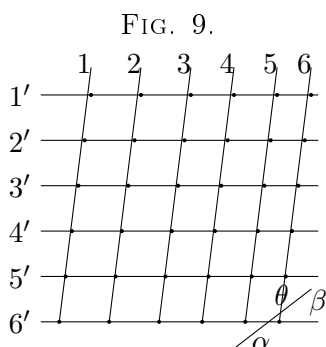
The Jacobian of any three conics is the locus of the points a , such that the polars of a in regard to the three conics meet in a point; and it is in general a cubic curve. Hence by what precedes it appears that the Jacobian of three circles is the cubic curve made up of the orthotomic circle and the line infinity.

13. *If five given lines have a common transversal, then taking the remaining transversal of each four of the given lines, show by statical considerations or otherwise that the five transversals have a common transversal.*

Consider the line $6'$ (fig. 9) meeting each of the lines 1, 2, 3, 4, 5, and take

- | | | |
|------|------------------------------|---------------|
| $1'$ | the remaining transversal of | (2, 3, 4, 5), |
| $2'$ | the remaining transversal of | (3, 4, 5, 1), |
| $3'$ | the remaining transversal of | (4, 5, 1, 2), |
| $4'$ | the remaining transversal of | (5, 1, 2, 3), |
| $5'$ | the remaining transversal of | (1, 2, 3, 4); |

it is to be shown that the lines $1', 2', 3', 4', 5'$ have a common transversal 6.



Consider the line 6 as a line determined so as to meet the lines $2', 3', 4'$ and $5'$; and take any line θ meeting each of the lines 6, $6'$; since the six lines 1, 2, 3, 4, 5, θ have a common transversal $6'$, then considering the lines as fixed lines in a solid body, it is possible to find along the lines 1, 2, 3, 4, 5, θ forces which will be in equilibrium. Suppose for a moment that the line 6, determined as above, does not meet the line $1'$; then we have the lines $1', 2', 3', 4', 5', \theta$ and 6 such that the line 6 meeting the lines $2', 3', 4', 5'$, and θ , does not meet the line $1'$. The six lines $1', 2', 3', 4', 5'$, and θ would be independent lines, such that there do not exist along them forces in equilibrium, and a force acting in any line whatever may be resolved

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into forces acting along the lines $1', 2', 3', 4', 5'$, and $6'$. Hence the original forces along the lines 1, 2, 3, 4, 5 can be each of them so resolved; and combining with the resolved forces the original force along θ , we have a system of forces along the lines $1', 2', 3', 4', 5', \theta$ in equilibrium with each other; which is impossible if the lines in question are related as above; that is, the line 6 meeting the lines $2', 3', 4', 5', \theta$ must also meet the line $1'$; which is the required theorem.

The statical principles assumed in the above demonstration are as follows:

- (1) Six lines may be such that there exist along them forces in equilibrium; or say, for shortness, the six lines may be in involution.
- (2) For any seven lines, no six or any less number of which are in involution, there exist along the lines forces in equilibrium; or, what is the same thing, given any six lines not in involution, and a seventh line; then a force along the seventh line may be resolved into forces along the six lines.
- (3) Six lines which have a common transversal are in involution (remark in passing that it is not conversely true that if the six lines are in involution they have a common transversal), but if five of the six lines have a common transversal not meeting the sixth line, then the six lines are not in involution.

14. In the theory of the variation of the arbitrary constants of a mechanical problem, state and explain the results obtained by Lagrange and Poisson respectively; and point out the peculiar advantage of Poisson's theory, in regard to the consequences which follow from the coefficients of his formulæ being independent of the time.

In the theory of the variation, of the arbitrary constants of a mechanical problem, it is assumed that the forces consist of principal and disturbing forces, each of them depending on a force function, say there is a principal force function and a disturbing function; (the assumption of a force function however is regard to the disturbing forces, though usual and convenient, is not essential); and that, when the disturbing forces are neglected, or say in the undisturbed problem, the equations of motion can be completely integrated; the theory consists herein that the same integral equations, taking the arbitrary constants to be variable, may be made to satisfy the disturbed equations of motion.

Suppose that the number of the coordinates $x, y, \dots$ is $= p$, then since the differential equations are of the second order, the number of arbitrary constants $(a, b, c, \dots)$ will be $= 2p$. In Lagrange's solution it is assumed that the coordinates $x, y, \dots$ (and consequently also the derived functions $x', y', \dots$) are each of them given in terms of t and the $2p$ constants; the disturbing function Ω is given in the same form; and the expressions for the variations $\frac{da}{dt}$, &c. are obtained by the solution of a system of linear equations

$$\frac{dR}{da} = (a, b) \frac{db}{dt} + (a, c) \frac{dc}{dt} + \&c.,$$

where the coefficients (a, b) are functions involving $\frac{dx}{da}, \frac{dx'}{da}$, &c.; these coefficients are

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time period functions of the $2p$ constants and of the time; but it is a result of the theory that they are in fact functions of the constants only, the time disappearing of itself.

In Poisson's theory it is assumed that the integrals of the undisturbed problem are obtained in the form

$$a = \phi(t, x, y, \dots, x', y', \dots), \text{ \&c.},$$

viz. that each constant is given in terms of the coordinates, their derived functions, and the time (so that, if all the integrals are known, this is equivalent to Lagrange's assumption). The expressions for the variations of the constants are given in the form

$$\frac{da}{dt} = (a, b) \frac{d\Omega}{db} + (a, c) \frac{d\Omega}{dc} + \text{\&c.},$$

where each coefficient (a, b) is given in the first instance expressed as functions of the coordinates $x, y, \dots$, the disturbing function Ω , &c. and the time; but it is a result of the theory that the coefficients (a, b) , &c. are really constant; viz. that, if $x, y, \dots, x', y', \dots$, are expressed in terms of the constants $(a, b, c, \dots)$ and of t , then the coefficient becomes a function of the constants $a, b, c, \dots$ only, the time disappearing of itself.

The transformation of the values of any coefficient $[a, b]$ requires only the knowledge of the expressions of a, b in terms of $x, y, \dots, x', y', \dots$, or we may say, the knowledge of the two integrals a, b . We thence obtain the expression of $[a, b]$ as a function of $x, y, \dots, x', y', \dots, t$, calling it, we have $\{a, b\} = \phi(t, x, y, \dots, x', y', \dots)$. But, in every transformation $[a, b]$ of the equations of motion of the undisturbed problem. It may happen that the value of $\{a, b\}$ is found to be a constant; in such case the constants of the values of the constants of the integrals all depend on the variables and the time. In every case, however, of the supposition that $\{a, b\}$ is independent of the time, leads to certain consequences analogous to this in Lagrange's theory.

15. Write a short dissertation on the transformation of rectangular coordinates in space of three dimensions, and in particular explain under what restriction it is true that two sets of rectangular axes, such that one system can by means of a rotation be made to coincide with the other, are conventionally said to be of the same kind, and that two sets of rectangular axes, such that the position of this axis and the second of the rotation cannot be made to coincide with each other, are of opposite kinds.

Two sets of rectangular axes about the same origin (each axis considered as it a line extending in the opposite sense from the origin) are said to be of the same kind if any one set can by displacement and rotation, be made to coincide with the other; in any other case they are of opposite kinds. It is easy to see that, if any one of the axes of the one set has been brought to coincide with the corresponding axis of the second set, then the other two axes will coincide with the corresponding axes,

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either the axis of z will coincide with that of z , or it will be in the opposite direction; in the former case the two sets are, in the latter case of the opposite kind, displacements one of the other. The restriction referred to in the question, is that the two sets shall be displacements one of the other.

In the problem of transformation, the two axis z axes, if not displacements, can always be made so by simply reversing the direction of one of the axes (writing for example x for x); and there is thus no real loss of generality in considering the two sets as displacements the one of the other; and it is in general convenient to make this assumption.

The transformation between two sets of rectangular axes is at once given by the diagram

	x	y	z
x'	a	β	γ
y'	a'	β'	γ'
z'	a''	β''	γ''

where a is the cosine of the inclination of the axis x , x_1 , and so for the rest of the nine quantities. The relation between the two sets of axes is obtained at pleasure by reading the diagram horizontally, $x = ax + \beta y + \gamma z$, &c. or by reading it vertically, $x = ax + a'y_1 + a''z'$, &c.

We must have identically $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, and we thus obtain the two equivalent sets of equations

$$\begin{array}{ll}
 a^2 + \beta^2 + \gamma^2 = 1, & a^2 + a'^2 + a''^2 = 1, \\
 a'^2 + \beta'^2 + \gamma'^2 = 1, & \beta^2 + \beta'^2 + \beta''^2 = 1, \\
 a''^2 + \beta''^2 + \gamma''^2 = 1, & \gamma^2 + \gamma'^2 + \gamma''^2 = 1, \\
 a''a' + \beta''\beta' + \gamma''\gamma' = 0, & \beta\gamma + \beta'\gamma' + \beta''\gamma'' = 0, \\
 a'a + \beta'\beta + \gamma'\gamma = 0, & a\gamma + a'\gamma' + a''\gamma'' = 0, \\
 aa' + \beta\beta' + \gamma\gamma' = 0, & a\beta + a'\beta' + a''\beta'' = 0.
 \end{array}$$

Either set leads to the relation

$$\begin{vmatrix} a, & \beta, & \gamma \\ a', & \beta', & \gamma' \\ a'', & \beta'', & \gamma'' \end{vmatrix} = \pm 1, \quad \text{consequently} \quad \begin{vmatrix} a', & \beta', & \gamma' \\ a'', & \beta'', & \gamma'' \\ a, & \beta, & \gamma \end{vmatrix} = \pm 1.$$

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the distinction between the two cases $+1, -1$ being that above explained; viz. if the axes are not displacements the one of the other, the sign is $+$; if they are, (and in all that follows this is assumed to be the case) then the sign is $-$. The equations give further $a = \beta'\gamma' - \beta''\gamma''$, &c. (six equations).

It is easy to see geometrically that, as stated in the question, two sets of axes (being so determined displacements the one of the other) can be made to coincide by means of a rotation of either axis through a certain axis; moreover, if the position of the axis itself is not altered by a, b, c, a and the angle of rotation, we must have

$$(b - a)x - \beta y + \gamma z = 0, \quad y(c - 1) = 0,$$

$$e'a + (\beta' - 1)y + \gamma'z = 0, \quad \&c.,$$

$$e''a + \beta''(b - 1)z = 0,$$

equations which must be equivalent to two equations; we in fact have

$$a : b : c : (a + b + 1)(a + b + 1)(c - 1) = \beta' - \gamma''(a - 1)(b + 1)(c + 1) = (1 - a)\gamma' - \beta'(c + 1),$$

as is easily verified by means of the foregoing relations between the coefficients. Any two of the three equations will then determine the ratios $a : b : c$; taking the second and third, we have

$$a : b : c = c(1 - b) - \alpha y(b - 1)\beta\gamma' : \beta'\gamma' + a'(1 - c)\gamma' + (1 - a)(1 + a)\gamma'' - 1)$$

reducible to

$$a : b : c = 1 + a - \beta'\gamma' : \beta - \alpha' : \gamma' + (1 - a)\gamma' - \beta'(c + 1)$$

and treating in the same way the third and first and second and first sets of equations the system of formulæ is

$$\begin{aligned} a : b : c &= 1 + a - \beta'\gamma' : \beta + \alpha' : \gamma' + a \\ &= a' : 1 + \beta' - a - \gamma'' \cdot y\beta + \gamma' \\ &= a' + a' : \gamma' + \alpha + a' : 1 + \gamma'' - a - \beta \end{aligned}$$

equations equivalent to each other; they determine the position of the axis in question, or resultant axis. The foregoing equations may also be written

$$a' : b' : c' : 1 = a + \beta' + \gamma' : 1 - a + \beta' + \gamma' : 1 + a - \beta' + \gamma' : 1 + a + \beta' - \gamma',$$

and hence, if A, B, C , be the inclinations of the resultant axis to the axes of x, y , and z , respectively, we have

$$\cos^2 A : \frac{1 - a + \beta + \gamma}{1 + a + \beta + \gamma}.$$

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and thence also

$$\begin{aligned}\sin^2 A &= \frac{2(1+a)}{3-a-\beta-\gamma}, \\ \cos 2A &= \frac{-1+3a-\beta-\gamma}{3-a-\beta-\gamma}, \\ a &= \cos 2A \frac{(1-a)(1+a+\beta+\gamma)+2a}{3-a-\beta-\gamma}.\end{aligned}$$

Let θ be the amount of the rotation about the resultant axis; we have a spherical triangle the two sides whereof are A, A , the included angle θ , and the opposite side $\cos^{-1} a$; that is, we have

$$\cos \theta = \frac{a - \cos^2 A}{\sin^2 A},$$

and thence

$$\begin{aligned}4 \cos^2 \frac{1}{2} \theta &= 2(1 + \cos \theta) = \frac{2(a - \cos 2A)}{\sin^2 A}, \\ &= 1 + a + \beta + \gamma,\end{aligned}$$

that is, the amount of the rotation is given by the formula

$$4 \cos^2 \frac{1}{2} \theta = 1 + a + \beta + \gamma.$$

The nine cosines a, β, γ , &c. may be expressed in terms of the inclinations A, B, C , and the rotation θ , or putting λ, μ, ν equal to $\tan \frac{1}{2} \theta \cos A, \tan \frac{1}{2} \theta \cos B, \tan \frac{1}{2} \theta \cos C$, in terms of the three quantities λ, μ, ν ; but the resulting formulæ for the transformation of coordinates can be more readily obtained by other methods.

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Solution of a Senate-House Problem.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. V. (1869), pp. 38-87.]

THE Problem, proposed January 7, 1869, is "If θ_1 and θ_2 are two values of θ which satisfy the equation

$$1 + \frac{\cos \theta_1 \cos \theta}{a^2} + \frac{\sin \theta_1 \sin \theta}{b^2} = 0,$$

show that θ_1 and θ_2 , if substituted for θ and ϕ in this equation, will satisfy it."

That is, writing

$$\frac{\cos \theta_1 \cos \theta}{a^2} + \frac{\sin \theta_1 \sin \theta}{b^2} + 1 = 0,$$

$$\frac{\cos \theta_1 \cos \phi}{a^2} + \frac{\sin \theta_1 \sin \phi}{b^2} + 1 = 0,$$

where $a^2 + b^2 = 1$, it is to be shown that

$$\frac{\cos \theta_1 \cos \theta_2}{a^2} + \frac{\sin \theta_1 \sin \theta_2}{b^2} + 1 = 0.$$

From the given equations, we have

$$\frac{\cos \phi}{a^2} : \frac{\sin \phi}{b^2} : 1 = \sin \theta_1 \sin \theta_2 : -\cos \theta_1 \cos \theta_2 : \sin(\theta_1 - \theta_2),$$

which is

$$= \cos \frac{1}{2}(\theta_1 + \theta_2) : -\sin \frac{1}{2}(\theta_1 + \theta_2) : 2 \cos \frac{1}{2}(\theta_1 - \theta_2).$$

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Whence eliminating ϕ , we have

$$a^2 \cos^2 \frac{1}{2}(\theta_1 + \theta_2) + b^2 \sin^2 \frac{1}{2}(\theta_1 + \theta_2) - a^2 b^2 \cos^2 \frac{1}{2}(\theta_1 - \theta_2) = 0,$$

that is,

$$a^2(1 + \cos(\theta_1 + \theta_2)) + b^2(1 - \cos(\theta_1 + \theta_2)) - (1 + \cos(\theta_1 - \theta_2)) = 0,$$

or, what is the same thing,

$$a^2 + b^2 - 1 + (a^2 - b^2) \cos(\theta_1 + \theta_2) - \cos(\theta_1 - \theta_2) = 0,$$

But from the equation $a^2 + b^2 = 1$, we have

$$a^2 + b^2 - 1 = -2a^2 b^2,$$

$$a^2 - b^2 - 1 = -2b^2,$$

$$-a^2 + b^2 - 1 = -2a^2,$$

and the equation is thus

$$\frac{\cos \theta_1 \cos \theta_2}{a^2} + \frac{\sin \theta_1 \sin \theta_2}{b^2} + 1 = 0,$$

which is the required equation.

Stopping at the result obtained previous to the use of the relation $a^2 + b^2 = 1$, but making some obvious substitutions in the formula, the theorem may be presented in a more general form as follows; viz. If we have

$$\frac{ax_1}{a^2} + \frac{by_1}{b^2} - 1 = 0,$$

$$\frac{ax_2}{a^2} + \frac{by_2}{b^2} - 1 = 0,$$

where

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{and} \quad \frac{x'^2}{a^2} + \frac{y'^2}{b^2} = 1,$$

then

$$\left(\frac{a}{a^2} + \frac{c}{b^2} - 1\right) \left(\frac{a'}{a^2} + \frac{c'}{b^2} - 1\right) + \left(\frac{c}{a^2} - \frac{a}{b^2}\right) \left(\frac{c'}{a^2} - \frac{a'}{b^2}\right) \left(1 - \frac{1}{a^2} - \frac{1}{b^2}\right) = 0,$$

a relation, the geometrical signification of which is: If (x, y) be a point on the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$,

and if the polar thereof in regard to the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meet the first-mentioned conic in the points (x_1, y_1) and (x_2, y_2) , then these points are harmonics in regard to the conic

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1\right) \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1\right) - \left(\frac{x^2}{a^2} - \frac{x^2}{b^2}\right)^2 \left(1 - \frac{1}{a^2} - \frac{1}{b^2}\right) = 0;$$

and since the theorem is projective, it is seen that three conics may be any conics whatever, the third conic being a conic having with the other two a common system of conjugate points.

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If to the ideas we write $a = b = c$; then the theorem is, if the polar of a point on the circle $a^2 + y^2 = a^2$ in regard to the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, meet the circle in two points, these are harmonics in regard to the conic $\frac{x^2}{a^2(a^2 + b^2 - 1)} + \frac{y^2}{b^2(a^2 + b^2 - 1)} = 1$; viz. if $a^2 + b^2 = 1$, the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, or if $a^2 = a^2$, $b^2 = b^2$, then the conic is the same similar to, but in the inverse ratio of the two points (x_1, y_1) , (x_2, y_2) harmonics in regard to the imaginary conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, but this is also an inverse form similar to the original conic, which is the theorem in question.

This last result will be similar to the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; viz. if $a^2 + b^2 = a$, then the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; if $a^2 = a^2$, $b^2 = b^2$, then $a^2 = a^2$, $b^2 = b^2 + b^2$; and the conic is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Considering the given conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ to be an ellipse, the first cone (a) is similar to the two points (x_1, y_1) , (x_2, y_2) harmonics in regard to the imaginary conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, viz. if $a^2 = a^2$, the conic of the right side of the equation is the same conic, but the points (x_1, y_1) , (x_2, y_2) are harmonics in regard to the imaginary conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, but this is also an inverse form similar to the original conic, which is the theorem in question.

The second case ($a^2 + b^2 = 1$) gives a real ellipse if b be negative; viz. this is, if for b we have the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, if $a^2 + b^2$, an obtuse-angled hyperbola, the circle $a^2 + y^2 - b^2$ which is the locus of the intersection of a pair of orthotomic tangents of the hyperbola, is consequently imaginary; but the eccentric orthotomic circle hence, on the limits of the theorem, must occur to have the opposite of polar of the property of a point on the circle in regard to the orthotomic hyperbola; viz. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = a^2$ also the theorem: "Given the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = a^2$ and the circle $a^2 + y^2 - a^2 + a^2$ (the concentric orthotomic circle of the imaginary circle which is the locus of the intersection of a pair of orthotomic tangents of the hyperbola); if the polar in regard to the hyperbola of a point on the circle meet the circle in two points Q , R , then these are harmonics in regard to the hyperbola."

Of course, if reality be disregarded, the two theorems are mere two of the cases of a conic generally; and observe, that in the first theorem the circle is the locus of the intersection of orthotomic tangents, and we have the opposite of the points Q , R in the second theorem the circle is the locus of the orthotomic circle of the cross, which the lines of in the cross of Q , R themselves.

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A “Smith’s Prize” Paper(*); Solutions.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. v. (1869), pp. 40-64.]

1. If $\sqrt{x - a^2 + (y + b)^2}$, $\sqrt{x - a^2 + (y - b)^2}$, a , β are respectively real and positive, show that in the equation $\sqrt{x - a^2 + (y - b)^2} = \sqrt{x - a^2 + (y + b)^2}$ is considered as representing a curve, the signs cannot either of them be assumed at pleasure to be + or be -; and distinguish the cases of the ellipse and the hyperbola.

Writing the equation in the form $\sqrt{x} \pm \sqrt{y} = c$, we find

$$\begin{aligned} 2(c\sqrt{r} \pm c\sqrt{r}) &= a + b - R, \\ 2(c\sqrt{r} + c\sqrt{r}) &= a + b - R, \\ 4cr &= (c + b)^2 - R^2, \\ 4cr &= (b - c)^2 - R^2, \\ 4cr &= (a + b)^2 - R^2, \end{aligned}$$

either of which equations is the rational equation of the curve (the equation being of the fourth order, inasmuch as $S - R$ is a linear function of the coordinates). But writing the rational equation under the form $a^2 + a^2 + 2kxy + xy(c - \beta)$, $i\beta(a - x)$, j , considered as representing a curve, the curve must either pass for any pair of values of x or y be positive, or in the opposite case be negative; that is, the equations $\sqrt{x} \pm \sqrt{y} = c$, will be in the one case real, in the opposite case imaginary. It is at once seen that, in either case, the original equation $\sqrt{x} \pm \sqrt{y} = c$ also represents the same algebraic value, found in any arbitrary form: but from this it is at once seen, that on a given form of $\sqrt{x} \pm \sqrt{y}$, a determinable value, fixed as soon as the values of x and the form of the curve, a determinable value, fixed as soon as the values of x and the form of the curve, a determinable value, fixed as soon as the values of x and the form of the curve, a determinable value, fixed as soon as the values of x and the form of the curve.

* Set by me for the Master of Trinity, A.D. 1869.

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point, but 2, 3, 4, 5, 6 are fixed points on the conic, then the pencil 1 (2, 3, 4, 5, 6) of lines from the point 1 to the four points 2, 3, 4, 5 has the same anharmonic ratio for all positions whatever of the variable point 1; which is the above-mentioned anharmonic property of the points of a conic.

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The “condition for an ellipse,” is that the given length c shall be greater than the distance $\sqrt{(a+b)^2 + (\beta-b)^2}$; between the two foci; for a hyperbola that it shall be less. In the former case, for any real point of the curve, it is obvious that the relation must be $\sqrt{x} + \sqrt{y} = c$; the latter case, the latter case, we must look at the actual position of the point.

It must be observed in regard to the ellipse, that, starting from the equation $\sqrt{x} + \sqrt{y} = c$, in finding the radical equation $\frac{x}{a^2} + \frac{y}{b^2} = 1$, $\frac{x}{a^2} - \frac{y}{b^2} = 1$, we have a perfect identical equation which, starting from $\sqrt{x} \pm \sqrt{y} = c$, signifies that we have

$$\sqrt{x} \pm \sqrt{y} = c, \text{ that is, } \sqrt{x} \pm \sqrt{y} = c, \text{ and } \sqrt{x} \pm \sqrt{y} = c,$$

but here, x being less than c , and the two signs are therefore each of them $+$, (c being positive), the two values verification applies to the hyperbola.

2. In a system of curves defined by an equation containing a variable parameter, investigate at any point the normal distance between two consecutive curves; and determine the form of the equation for a system of parallel curves.

Consider the system of curves $f(x, y, c) = 0$; then if the point x, y belongs to the curve $f(x, y, c) = 0$, and the point $x + dx, y + dy$ to the curve $f(x, y, c + dc) = 0$, we have

$$\frac{df}{dx} dx + \frac{df}{dy} dy + \frac{df}{dc} dc = 0,$$

and if the point $x + dx, y + dy$ be on the normal at (x, y) to the curve $f(x, y, c) = 0$, we have

$$dx : dy = \frac{df}{dx} : \frac{df}{dy},$$

or writing

$$dx = k \frac{df}{dx}, \quad dy = k \frac{df}{dy},$$

we have

$$k \left\{ \left(\frac{df}{dx} \right)^2 + \left(\frac{df}{dy} \right)^2 \right\} + \frac{df}{dc} dc = 0,$$

wherefore

$$k = -\frac{df}{dc} dc \div \left\{ \left(\frac{df}{dx} \right)^2 + \left(\frac{df}{dy} \right)^2 \right\},$$

and the normal distance at (x, y) of the curves c and $c + dc$ is $= \sqrt{(dx^2 + dy^2)}$, viz. it is

$$\begin{aligned} &= k \sqrt{\left\{ \left(\frac{df}{dx} \right)^2 + \left(\frac{df}{dy} \right)^2 \right\}}, \quad \text{or finally it is} \\ &= -\frac{df}{dc} dc \div \sqrt{\left\{ \left(\frac{df}{dx} \right)^2 + \left(\frac{df}{dy} \right)^2 \right\}}, \end{aligned}$$

where, if the distance in question be regarded as positive (that is, if we attend only to its absolute value), the sign is to be taken so that $\frac{df}{dc}$ shall be positive.

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If the question be given in the form $V - c = 0$ (V a function of (x, y)), we have the normal distance

$$= \pm dc \div \sqrt{\left\{ \left(\frac{dV}{dx} \right)^2 + \left(\frac{dV}{dy} \right)^2 \right\}}.$$

For parallel curves, the normal distance is everywhere the same, that is, $\left(\frac{df}{dx} \right)^2 + \left(\frac{df}{dy} \right)^2$ must have a constant value for all values of (x, y) satisfying the relation $V = c$, viz. we must have identically $\left(\frac{df}{dx} \right)^2 + \left(\frac{df}{dy} \right)^2 = \phi(V)$, ϕ arbitrary; a partial differential equation to be satisfied by V in order that the system $V = c$ may be the equation of a system of parallel curves. Assuming the equation to be satisfied for any particular form of ϕ , say put θ in place, then if C a function of V , such that $\left(\frac{dC}{dV} \right)^2 \phi(V) = 1$, and the equation $f(V) = \text{Const.}$ is the same thing as $V = \text{Const.}$, it follows that the equation of the system of parallel curves may be taken to be $V = c$, where

$$\left(\frac{dV}{dx} \right)^2 + \left(\frac{dV}{dy} \right)^2 = 1.$$

3. Two cannons (each free to recoil) differ only in weight and in the weight of the ball; and it is assumed that at any instant during the explosion the explosive force depends only on the space occupied by the vapour of the gunpowder: compare the emerging velocities of the balls; and also the emerging velocities of balls fired from the same cannon when it is free to recoil, and when it is absolutely fixed.

Consider a single cannon.

Take M for its mass, m for that of the ball, S , s for the spaces described, backwards by the cannon and forwards by the ball, at the time t during the explosion. Then by hypothesis the explosive force, forwards on the ball and backwards on the cannon, is a function of $s + S$, $= \phi(s + S)$ suppose, or we have

$$m \frac{d^2 s}{dt^2} = \phi(s + S),$$

$$M \frac{d^2 S}{dt^2} = \phi(s + S),$$

and thence

$$\frac{d^2(s + S)}{dt^2} = \left(\frac{1}{m} + \frac{1}{M} \right) \phi(s + S).$$

Multiplying each side by $2 \frac{d(s + S)}{dt}$, integrating from $s + S = 0$, to $s + S = a$, where a is the length of the tube, and observing that the initial velocities are each $= 0$, we find

$$\left\{ \frac{d(s + S)}{dt} \right\}^2 = 2 \left(\frac{1}{m} + \frac{1}{M} \right) \int_0^a \phi(u) du,$$